Variance-Swap Targets

Log-contract replication, the LQD closed form, and the source-PDE method

Vol-Fitter Technical Notes, No. 08

Vol-Fitter

July 3, 2026

Abstract

A variance-swap quote is one number per smile node — the market's fair var-swap volatility for an (underlying, T) — and it pins a global property of the smile that scattered option quotes constrain only weakly: the total risk-neutral variance, dominated by the wings. This note documents how the Vol-Fitter turns a var-swap quote into a soft calibration target shared by every model. The fair var-swap total variance is the model-free OTM log-contract replication; the penalty pulls the model's *own* fair var-swap toward the quote, in vol units, weighted as a chosen fraction of the node's option weight. Computing the model's var-swap cheaply is the crux: the generic replication re-solves implied variance on a grid every Jacobian column and would make a single fit take minutes, so LQD uses its *exact closed form* ($-2\,\mathbb{E}[X]$) and the local-vol surface uses a *source PDE* ($I(T) = g(0,1)$) with analytic sensitivities. A fresh run confirms the LQD closed form agrees with replication to 1.53 vol bps, the replication converges by a half-width of ~ 4, and a var-swap quote 25.97% pulls a smile whose natural var-swap is 24.47% up to 25.91% while keeping the option fit.

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1 What a variance swap pins

A variance swap pays the realized variance of the underlying to T against a fixed strike. Under a diffusion the fair strike equals the risk-neutral expected total variance, which the model-free Carr–Madan log-contract replication expresses as a portfolio of OTM options:

$$w_{\mathrm{vs}}(T) = \mathbb{E}^T\Big[\int_0^T \sigma_t^2 \,\mathrm{d}t\Big] = 2\int_0^\infty \frac{P(T,K)}{K^2}\mathrm{d}K_{\mathrm{put}} + 2\int^\infty \frac{C(T,K)}{K^2}\mathrm{d}K_{\mathrm{call}}, \tag{1}$$

the $1/K^2$ weighting placing most of the mass in the *wings*. This is exactly why a var-swap quote is informative: it constrains the total variance, a global moment that a handful of near-the-money option quotes barely touch. A smile can fit every quoted strike and still carry the wrong wing mass; the var-swap quote pins it.

Heuristic

Think of the var-swap as a single, very reliable "wing quote". Liquid options cluster near the money, where they say little about the tails; the var-swap is a traded instrument whose value is *dominated* by the tails (the $1/K^2$ weight). Adding it to the objective is the cheapest way to discipline the extrapolation the option quotes leave free — complementary to the Lee wing control of Note 09, which constrains the wing *slope* while the var-swap constrains the wing *mass*.

2 The replication and the penalty

In forward-normalized units ($F = 1$, $k = \log K$), (1) becomes a single integral over the normalized Black call $B(k, w(k))$,

$$w_{\mathrm{vs}} = 2\int_{-\infty}^{\infty} \Big[B(k, w(k)) + (e^k - 1)^-\Big] e^{-k}\,\mathrm{d}k, \tag{2}$$

evaluated by a trapezoid on $k \in [-H, H]$ with $H = 6.0$ and 801 points (the put leg adds $(e^k - 1)e^{-k}$ for $k < 0$). In code, (2) is a trapezoid over any model's $w(k)$ (matches production to 10^{-9}):

Listing 1: Model-free var-swap total variance by log-contract replication (2) (distilled from `calib/varswap.py`).

```
def varswap_total_variance(implied_w, half=6.0, n=801):    # implied_w(k) -> total variance
    k = np.linspace(-half, half, n)
    w = np.maximum(implied_w(k), 1e-12)
    integ = black_call(k, w)*np.exp(-k)                  # OTM call leg, discounted
    put = k < 0
    integ[put] += 1 - np.exp(-k[put])                    # (e^k - 1) e^{-k} on the put side
    return 2.0*np.trapezoid(integ, k)
```

The half-width is coarser than the diagnostics grid because this runs *inside* the optimizer loop; ± 6 captures the OTM mass to well under a basis point for any sane smile, as fig. 1 confirms — the fair var-swap vol has plateaued by a half-width of ~ 4.

The penalty is a single residual in *volatility* units, so it stacks directly onto the vol-space data residuals of SVI/SIV and the vega-normalized residuals of LQD/LV:

$$\boxed{r_{\text{vs}} = \sqrt{\omega_{\text{vs}}}\,(\sigma_{\text{vs}}^{\text{model}} - \sigma_{\text{vs}}^{\text{quote}}), \qquad \sigma_{\text{vs}} = \sqrt{w_{\text{vs}}/T}.} \tag{3}$$

The weight ω_{vs} is set to a chosen fraction (`varSwapWeightPct`, default 10%) of the node's summed option-quote weight, so the var-swap contributes a controlled share of the total objective rather than an absolute, scale-dependent amount.

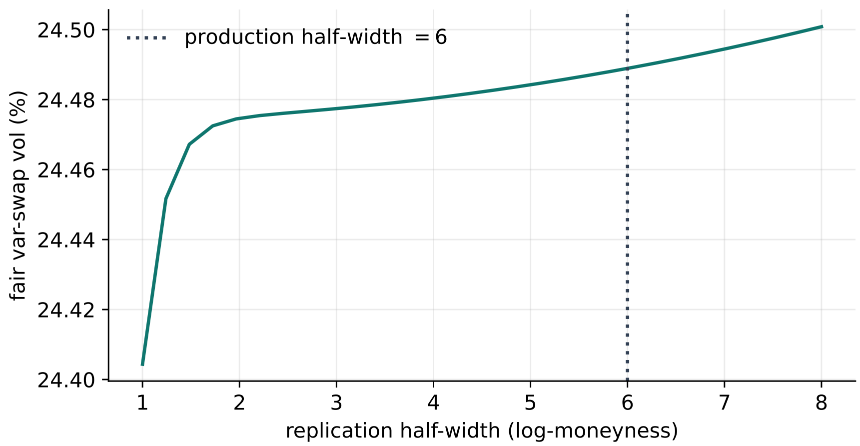

Figure 1: Fair var-swap volatility vs the replication half-width. The integral has converged well before the production half-width of 6.

3 Computing the model's var-swap cheaply

The penalty needs the *model's* fair var-swap at every optimizer iteration. Doing that by the generic replication (2) is a trap: it calls the model's implied_w(k) on ~ 800 points, and for LQD each such point is a Black inversion of a quadrature — so a single Jacobian column triggers thousands of root finds and the fit takes minutes. Each model therefore supplies its *own* cheap var-swap.

3.1 LQD: the exact closed form

For LQD the var-swap is a one-line consequence of the log-contract identity $\mathbb{E}[-2X] = -2\int_0^1 Q(u)\,\mathrm{d}u$. In the logit coordinate this is a single quadrature against the *same* grid the slice already built,

$$w_{\text{vs}}^{\text{LQD}} = -2\,\mathbb{E}[X] = -2\int_{-\infty}^{\infty} Q(z)\,\Lambda(z)(1-\Lambda(z))\,\mathrm{d}z, \tag{4}$$

whose integrand decays like $z\,e^{-|z|}$. No inversion, no extra grid — it is essentially free. Table 1 confirms it agrees with the generic replication to 1.53 vol bps, validating both.

3.2 SVI and SIV: arithmetic curves

For SVI and Multi-Core SIV the total variance $w(k)$ is closed-form arithmetic (a hyperbola, or a log-cosh sum), so the replication (2) is cheap directly — no inversion is needed because $w(k)$ is evaluated, not solved.

3.3 Local volatility: the source PDE

The piecewise-affine surface has no arithmetic $w(k)$; pricing each k means a PDE solve. The *source-PDE* method (Note 04) instead reads the var-swap off one backward parabolic solve,

$$\partial_t g + \tfrac{1}{2}\nu(t,x)x^2\partial_{xx}g + \nu = 0, \quad g(T,\cdot) = 0, \qquad I(T) = g(0,1), \tag{5}$$

with analytic sensitivities $\partial I/\partial\theta$ from the same multi-RHS machinery as the value solve, and — unlike the replication — no dependence on the truncation $x_{\max}$. It is gated by `varSwapMethod` (default `static` replication; the source PDE is the faster alternative).

4 Worked example

Figure 2 fits an LQD slice to an index-like smile whose *natural* fair var-swap is 24.47%, then refits it with a var-swap quote of 25.97% (1.5 vol points higher) at 10%-class weight. The targeted fit lifts its fair var-swap to 25.91% — almost all the way to the quote — while still threading the option quotes: the var-swap adds wing mass the near-the-money options did not constrain. Table 1 collects the numbers.

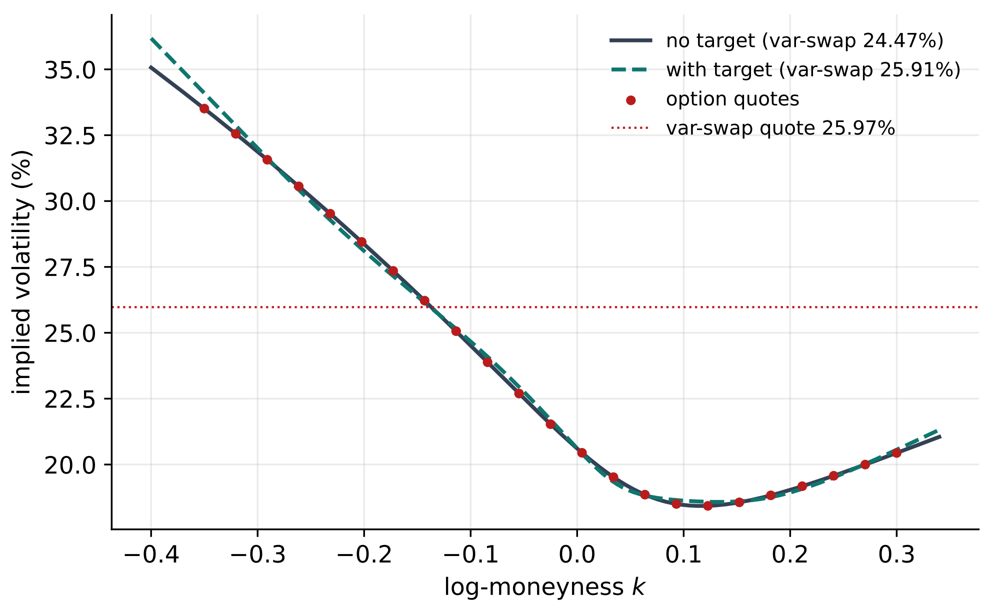

Figure 2: An LQD fit with (dashed) and without (solid) a var-swap target above the smile's natural level. The target lifts the fair var-swap toward the quote by adding wing mass while keeping the option fit.

Table 1: Var-swap numbers (fresh run): closed form vs replication, and the targeted pull.

Quantity	Value
LQD closed-form var-swap vol	24.47%
Replication var-swap vol	24.49%
Closed-form vs replication agreement	1.53 vol bp
Var-swap quote (target)	25.97%
Targeted-fit fair var-swap	25.91%

5 What is original here

The log-contract replication is Carr–Madan; the contributions are practical. The *per-model cheap var-swap* — LQD's closed form (4), SVI/SIV's arithmetic curves, and the LV source PDE (5) with analytic sensitivities — is what makes a var-swap target affordable *inside* the optimizer loop rather than as a slow post-hoc check; and the *relative* weighting (a fraction of the node's option weight) keeps the target scale-free across nodes with very different quote counts. Together they make one extra global constraint cost essentially nothing.

A Hyperparameter atlas

Table 2: Variance-swap hyperparameters.

Knob	Default	Role
Surfaced		
`varSwapEnabled`	off	Master toggle for the var-swap target.
`varSwapWeightPct`	10%	Var-swap weight as a fraction of node option weight.
`varSwapMethod` (LV)	`static`	Replication or source-PDE pricer.
Hidden		
`VS_HALF_WIDTH` H	6.0	Replication half-width in log-moneyness.
`VS_POINTS`	801	Replication trapezoid points.
`_W_FLOOR`	10^{-12}	Total-variance floor in the integrand.

Performance. The LQD closed form is one quadrature on the existing grid (free); the replication is ~ 800 arithmetic evaluations for SVI/SIV (negligible); the LV source PDE is one extra backward solve with analytic sensitivities. Using the generic replication on an LQD curve — ~ 800 Black inversions per Jacobian column — would turn a millisecond fit into minutes, which is the whole reason for the per-model routes of section 3.

B Reference implementation: the LQD closed form

For LQD the fair var-swap total variance is $-2\,\mathbb{E}[X]$ with no replication integral and no Black inversions: the log contract is exactly the mean log-return, and the quadrature grid already carries

$Q(z)$ and $u(1-u)$ (Note 01). Agrees with the generic replication to 1.53 vol bps and with production to 10^{-12}.

Listing 2: The LQD closed-form var-swap strike, the mean log-return (distilled from `models/lqd/quadrature.py`).

```
def var_swap_strike(z, q_z, u):            # z: logit grid, q_z = Q(z), u = expit(z)
    return -2.0 * np.trapezoid(q_z * u * (1 - u), z)   # = -2 E[X], the log contract
```

References

[1] P. Carr and D. Madan. Towards a theory of volatility trading. In *Volatility*, Risk Books, 1998.

[2] K. Demeterfi, E. Derman, M. Kamal and J. Zou. A guide to volatility and variance swaps. *J. Derivatives*, 6(4):9–32, 1999.

[3] J. Gatheral. *The Volatility Surface*. Wiley, 2006.